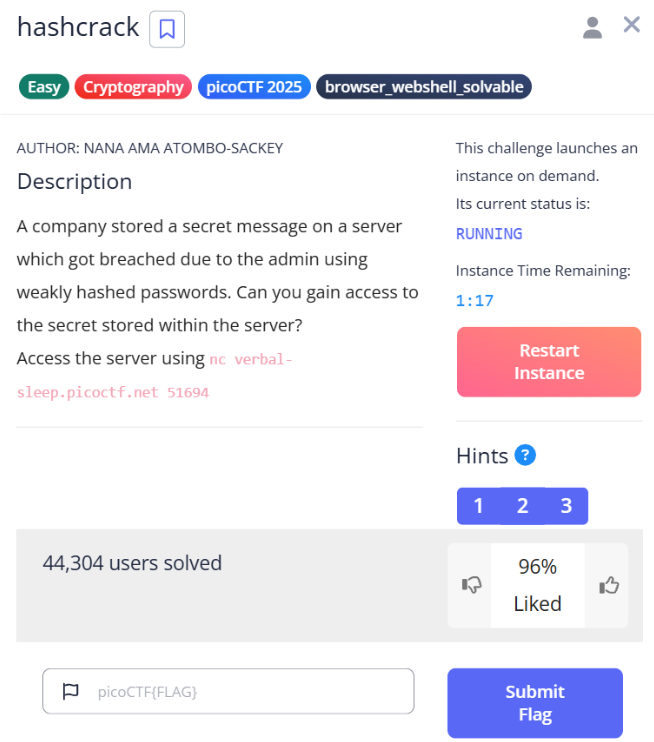
HERNANDO, PANTINE B. **BSIT-3**

CHALLENGE 1 – CRYPTOGRAPHY



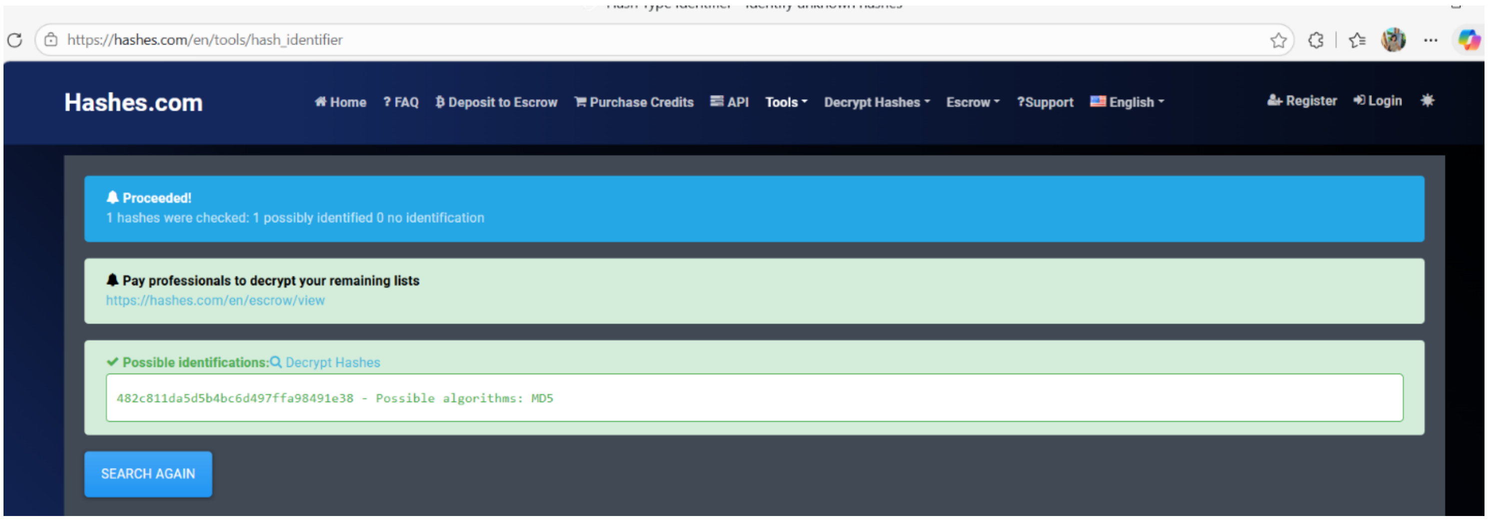
Step-by-step solution

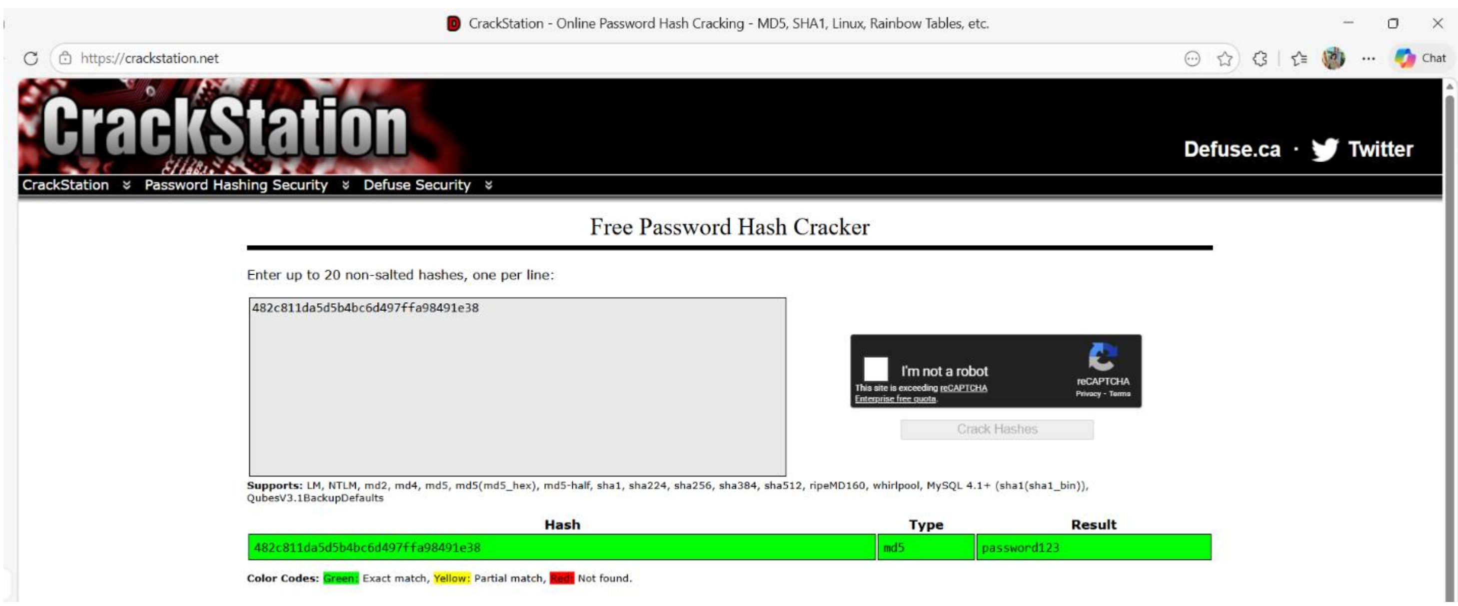
1. Initial connection

- Access the challenge server via the provided nc (netcat) command in the web shell.

nc verbal-sleep.picoctf.net 51694

```
We have identified a hash: 482c811da5d5b4bc6d497ffa98491e38
Enter the password for identified hash: password123
Correct! You've cracked the MD5 hash with no secret found!
```

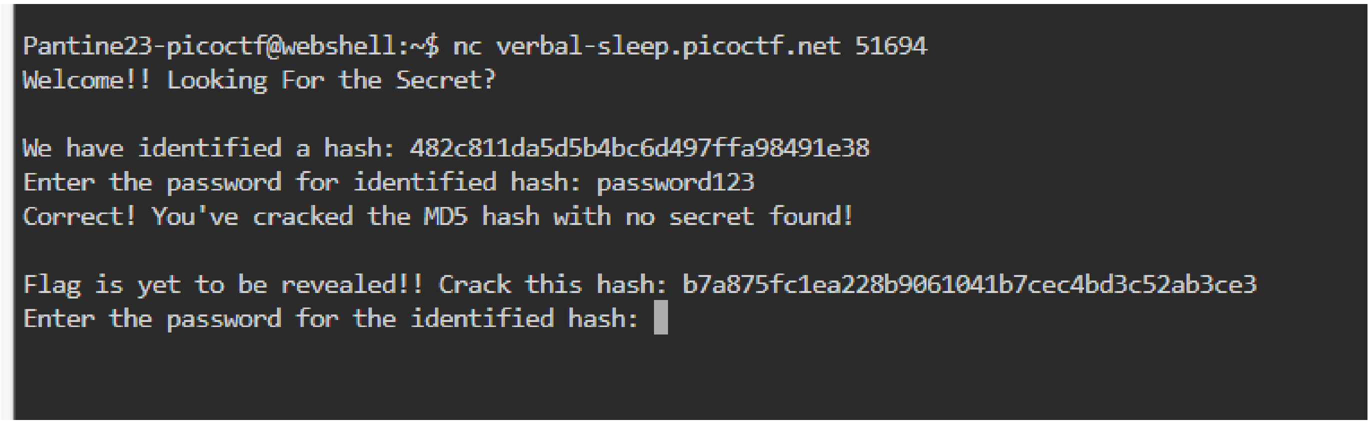


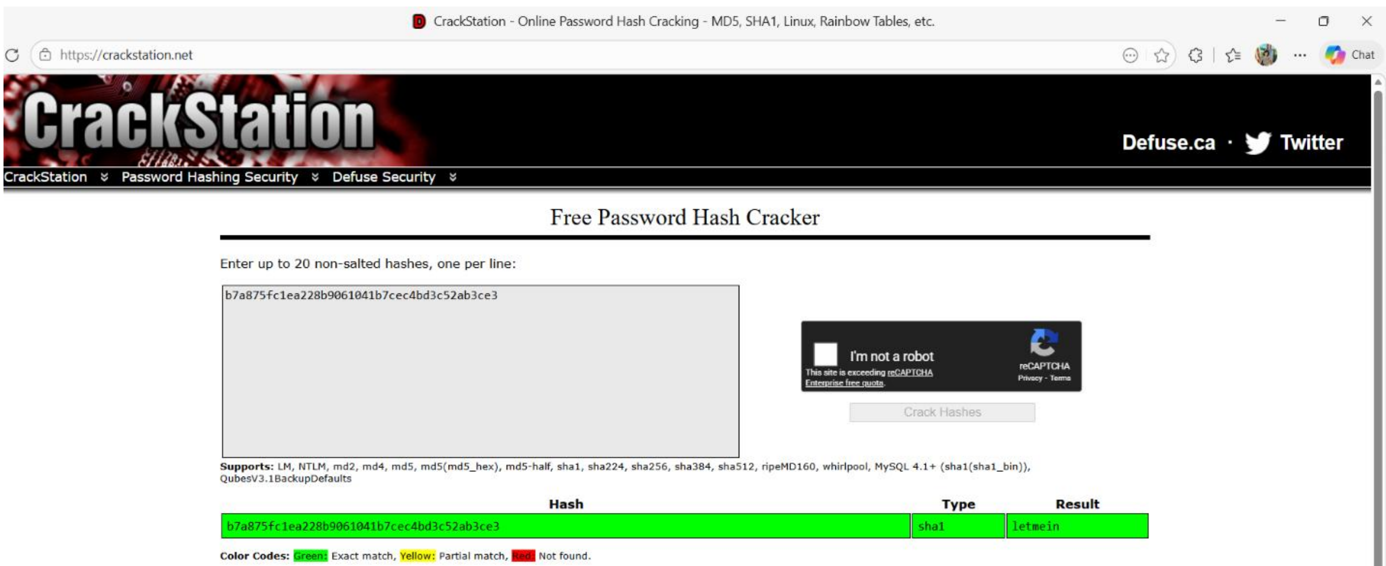


2. *Cracking the MD5 Hash*

The server presents the first hash: 482c811da5d5b4bc6d497ffa98491e38.

- **Tool:** CrackStation or Hashes.com.
- **Result:** The password for this MD5 hash is **password123**.
- **Outcome:** The server confirms it is correct but states no secret was found yet.

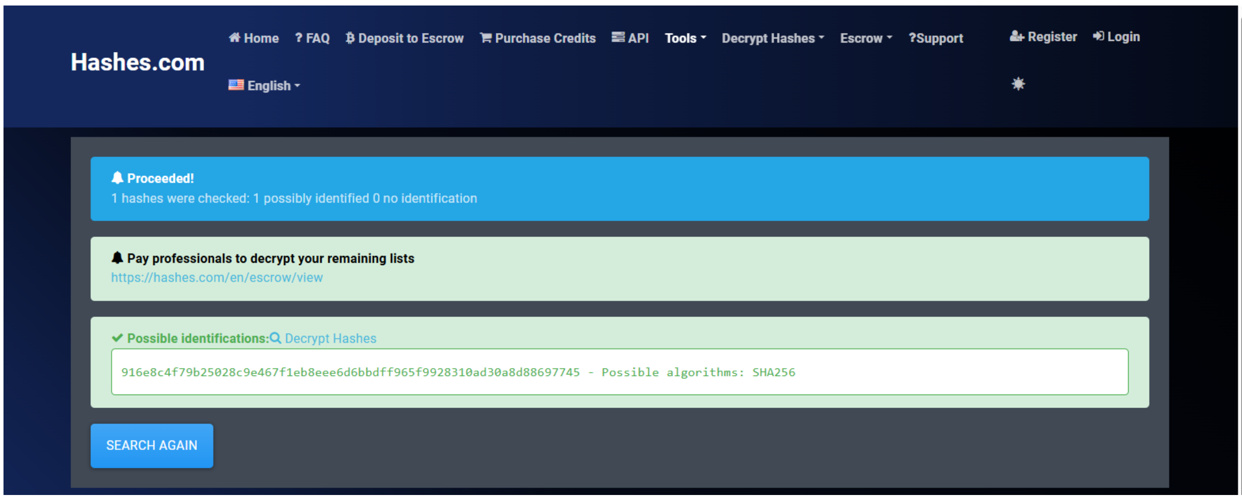


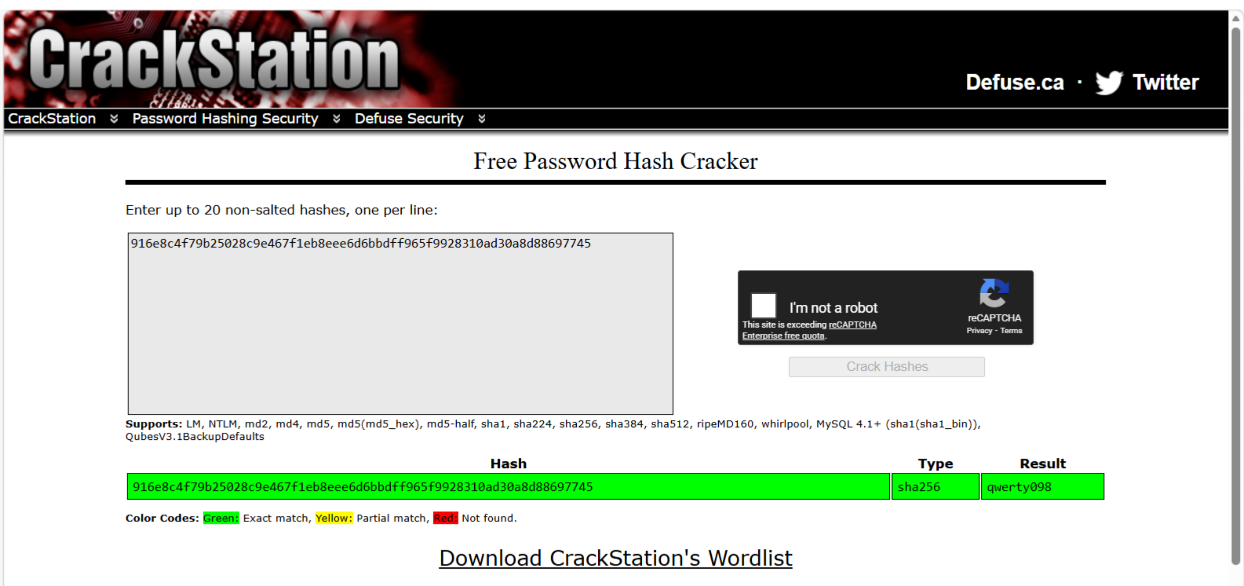


3. ***Cracking the SHA-1 Hash***
   The server then provides a second hash: b7a875fc1ea228b9061041b7cec4bd3c52ab3ce3.
   - **Tool:** CrackStation.
   - **Type:** Identified as **SHA-1**.
   - **Result:** The password for this hash is **letmein**.
   - **Outcome:** The server confirms the SHA-1 crack but indicates you are "almost there".

```
Flag is yet to be revealed!! Crack this hash: b7a875fc1ea228b9061041b7cec4bd3c52ab3ce3
Enter the password for the identified hash: letmein
Correct! You've cracked the SHA-1 hash with no secret found!

Almost there!! Crack this hash: 916e8c4f79b25028c9e467f1eb8eee6d6bbdff965f9928310ad30a8d88697745
Enter the password for the identified hash: qwerty098
Correct! You've cracked the SHA-256 hash with a secret found.
The flag is: picoCTF{UseStr0nG_h@shEs_&PaSswDs!_5b836723}
```





4. Cracking the SHA-256 Hash

The final hash provided is: 916e8c4f79b25028c9e467f1eb8eee6d6bbdff965f9928310ad30a8d88697745.

- **Tool:** CrackStation or Hashes.com.

- **Type:** Identified as **SHA-256**.
- **Result:** The password for this hash is **qwerty098**.

Note: Some search results on CrackStation may show "qwerty" for similar hashes, but the server specifically accepts **qwerty098** to progress.

```
Pantine23-picoctf@webshell:~$ nc verbal-sleep.picoctf.net 55589
Welcome!! Looking For the Secret?

We have identified a hash: 482c811da5d5b4bc6d497ffa98491e38
Enter the password for identified hash: passsword123
Incorrect. Goodbye.
no
Pantine23-picoctf@webshell:~$ nc verbal-sleep.picoctf.net 55589
Welcome!! Looking For the Secret?

We have identified a hash: 482c811da5d5b4bc6d497ffa98491e38
Enter the password for identified hash: password123
Correct! You've cracked the MD5 hash with no secret found!

Flag is yet to be revealed!! Crack this hash: b7a875fc1ea228b9061041b7cec4bd3c52ab3ce3
Enter the password for the identified hash: letmein
Correct! You've cracked the SHA-1 hash with no secret found!

Almost there!! Crack this hash: 916e8c4f79b25028c9e467f1eb8eee6d6bbdff965f9928310ad30a8d88697745
Enter the password for the identified hash: qwerty098
Correct! You've cracked the SHA-256 hash with a secret found.
The flag is: picoCTF{UseStr0nG_h@shEs_&PaSswDs!_5b836723}
```

SHORT REFLECTION:

This analysis clearly illustrates that weakly hashed passwords increase the likelihood of data breaches. The use of common words such as "password123" and "letmein" by system administrators makes it easy for attackers to breach systems through publicly available rainbow tables and password cracking tools such as CrackStation and Hashes.com. The transition from MD5 to SHA-1 and then to SHA-256 clearly shows that even mathematically superior password cracking methods are useless when weakly hashed passwords are used. As the flag clearly shows, the best way to protect against these attacks is to implement strong password hashing and complex passwords.